

Abstract

Mechanically updated Semi-Automatic Tile Installing Machine is designed to enhance the precision, accuracy and efficiency of floor tile construction and cost effectiveness of manual labor-intensive construction. A new tile application can be met with uneven application of glue, poor lining and lengthy procedures. This work meets these challenges by designing and implementing major mechanical components into a small, easy-to-use machine.

The system will have structural rigid frame to handle the required structural feeds, an adhesive spreading system which will ensure the full spread over of adhesive, tile placement arm which will be used to ensure tiles laid perfectly at the right position, a suction arm module which will aid in secure holding of tiles and a spirit levelling which will ensure constant and flat positioning of tile. SolidWorks was used to model each subsystem, and the sub systems assemblies were verified by motion analysis and Finite Element Analysis (FEA) critical parts that carry the load to ensure strength, dependability and safety under operating conditions.

The requirements were analyzed, concept was developed, the detailed 3D modelling and subsystem integration were made. Structural, torque, and adhesive roller engineering were calculated, to have realistic viability. All the solutions provided prove that the new machine would be able to provide accurate positioning of tiles, minimize installation mistakes, and improve productivity at a reasonable cost for small contractors and domestic users.

Finally, the project demonstrates that the problem of outdated conventional tile installation could be fixed with the help of the simple mechanical innovations which have been assisted with CAD-based and simulation tools. This machine offers cost effective semi-automation that offers higher precision, labor saving, and in future can be used in incorporation of smart sensors and even increased automation.

Keywords

Semi-automatic tile installing machine, CAD modeling, FEA analysis, adhesive spreading system, suction arm module, spirit levelling, construction automation, productivity improvement, cost-effective mechanization, small-scale contractors.

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Introduction

Background

Tiling is a sensitive process in the process of construction and interior finishing in terms of accuracy in the way they are laid, the uniformity in the adhesive use, consistency in the spacing and the levelness of the surfaces to which they are put on. This has traditionally been a labor intensive, and very skill-reliant process because it would require skilled and experienced tilers. Manual tile laying may encounter limitations like difficulty in placement, excessive adhesive, and irregularity in tile spacing that may affect looks and integrity of the surface that will be developed.

On small scale domestic projects or DIY installations, lack of skilled labor can lead to errors and the time taken to complete the installation process takes longer than expected. After the contractor, the procedure is recursive, labor-intensive and time-consuming. As the world embraces the importance of productivity and quality, the construction sector has been shifting towards mechanization and automation in bid to rationalize the construction process and one such aspect is tiling.

There are three tiers of solutions used, simple manual aids (suction lifters, tile vibration tools) as well as complete robotic settings. Suction and vibration handheld semi-automatic tools guarantee even adhesive coverage and help to enhance quality without necessarily eliminating manual application. At the premium end, autonomous applications such as the Partner Robotics P900 and BDR R-19 achieve speeds of one tile every 40 seconds using robotic arm, suction shutter and AI-generated alignment mechanisms. [1] Such systems have enhanced capabilities in the form of automated mortar, SLAM navigation and visual alignment sensors.

There are disadvantages associated with the current solutions of robotic tiling. They tend to be costly (thousands of dollars) bulky (up to 800 kg) and are most suited to large-scale commercial applications. [2] A small-scale contractor and a home consumer still use manual or semi-automatic type tools owing to the issue of cost, portability and ease of use. This opens a market with semi-automated inexpensive, lightweight and practical robots that can help install the tiles accurately and without the cost and complexity of full-scale robots.

Aim

The aim of this project is to design and validate a **semi-automatic tile installing machine** that improves tile laying accuracy, consistency, and efficiency for small-scale and domestic applications, while remaining affordable and user-friendly.

Objectives

To develop CAD models of all major subsystems, including the frame, adhesive spreading system, tile placement arm, suction arm module, and spirit levelling system.

To automate adhesive spreading and tile placement to ensure consistent bonding and accurate positioning.

To ensure accuracy of spacing and leveling using guide rails and integrated leveling mechanisms.

To validate the design with Finite Element Analysis (FEA) for structural integrity, stress distribution, and motion performance.

To prepare a complete production drawing package and technical documentation for potential prototyping and manufacturing.

Project Specification

Scope

The project is more design related, analysis, and validation which is being done through CAD modelling and simulations. Physical prototyping is outside the timeframe, but it is recommended that it should be done in future.

Subsystems

Frame, adhesive spreading system, tile placement arm, suction arm module, and spirit levelling system.

Target Users

DIY enthusiasts, domestic users, and small contractors requiring professional-quality results without specialized skills.

Design Approach

Low-cost, modular construction using commonly available materials and standard fasteners. Emphasis on portability, ease of operation, and integration of semi-automated mechanical functions.

Limitations

Large-scale industrial applications and full robotic automation are beyond the scope of this project.

Literature Review

Traditional Tile Installation Challenges

Installation of floor tiles is a usual manual, labor-intensive process which requires accuracy and performance. Employees should add mortar (tile glue), and lay each tile into proper positioning and spacing, and apply enough mortar so that the surface is level (avoiding uneven edges). Applying a level, smooth surface is dependent on experience and close methods on the part of the installer. With manual approaches the position and level of tiles can exhibit inconsistencies which cause defects that affect the quality of the finish of a room. In addition, such work is physically very demanding since installers spend a lot of time bending or kneeling to handle heavy tiles and construction materials. This repetitious stress leads to employee exhaustion and injuries, and provides motive images, deterring many prospective employees from going into the trade. Meanwhile, construction sectors around the globe are experiencing skilled tiler deficits, as well as increasing labor expenses. These difficulties justify the importance of technological advances in tile installation and improvements in tile installation to increase precision, productivity, and ergonomics. [3]

Researchers have singled out accuracy (proper destination and alignment, not-left behind, and fascinated leveling) as one of the main issues in the automation of tile installation. Even with the use of tiling aids (flat tiles or laser guides), one will often not be able to guarantee that each tile is fitting flush with the other tile by hand. Research indicates that perfect tile lay and leveling are hard to achieve thus not enabling complete automation to be achieved in actual practice. Conventional tiles are also not too productive. Man is typically able to lay down the approximately ten of square meters per day. By contrast, an effectively engineered roboticized solution has the potential to tile more quickly and consistently, resolving the speed and quality problems. These have been the driving forces behind academic and commercial research on mechanized, automated or semi-automated tile-laying systems in recent decades.

Mechanical Design Strategies in Tile-Laying Machines

The mechanical tasks to pick up the tiles, apply adhesive, place the tile correctly, and provide a flat surface are to be solved to develop the machine to install tiles. Past research and product prototypes have experimented with a variety of design solutions to these functions.

- **Tile Placement Arms and Suction Grippers**

Robotic arm or automatic placing is used in most automated tiling systems, where the gripping force is usually suction. Vacuum suction cups enable a machine to raise and place tiles without damaging them, resembling how a person might use a suction tool.

As an example, a recent heavy-tile robot adopts an array of four suction cups on its end-effector to firmly hold large ceramic tiles. [4] A 6-DOF multi-axes robotic arm or parallel manipulator is required to provide the required motion range to move tiles off a stack and to the floor. A six-axes arm can align each tile perfectly, and some configurations include compliance (flexibility) in the gripper so that each tile may be pressed into place without breaking. This hand and machine compatible laying method provides fast and precise work without causing breakage either on the tile or to the neighboring one.

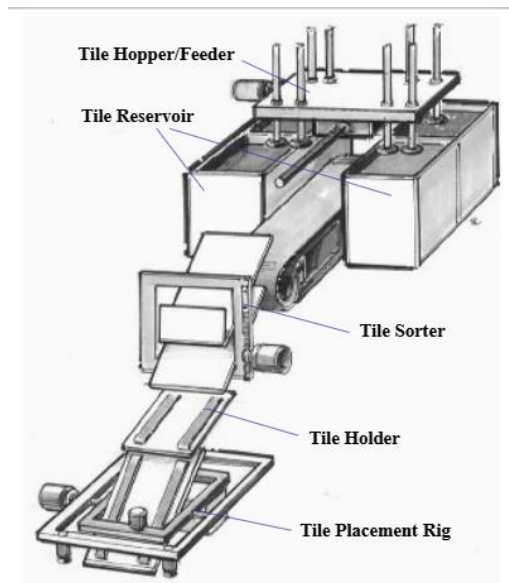


Figure 1: Current system

- **Adhesive Application Systems**

In addition to tiles, an installation machine should also deal with the tile adhesive (mortar). Multiple methods have been envisioned to remove the manual spreading of adhesives. Other systems are independently outfitted with a store of mortar/ glue and dispense it through pumps or extruders. One patent outlines a brushing mechanism within the robot which automatically has adhesive brushed onto each of the tiles prior to their placement. Such a device contains an extruder screw and motor to mete out a precisely applied layer of mortar on the underside of the tile without troweling. [5] Vibrating mortar trays or rollers have been used on other designs to produce a uniform distribution of adhesives on the substrate. Maintaining stabilized adhesive thickness is of paramount importance to both bond and levelness, something which the robotic method can maintain more accurately than manual application of ‘thinset’. Fully automated prototypes have also been developed that in-situ mix adhesives and feed them by hoses though these are complex and add weight to the system.

- **Leveling and Alignment Mechanisms**

To obtain level flooring, machines incorporate sensors and force control as they lay each tile. The typical tactic is to integrate a leveling mechanism or vibrating unit that imprints the tile to the subsurface of the mortar bed in an equal state. As an example, researchers have incorporated eccentric vibrators into the tile placement head - once a tile has been laid, a quick vibration assists the tile to settle and prevents the air gap, resulting in greater contact and leveling. Empirical tests demonstrate that this vibration makes a large difference in adhesive contact achieved and prevents the tiles being forced out of plane with subsequent rebound. One more is the alignment part; machine vision (cameras) or laser sensors are used to sense the tile edges and neighboring tiles, so the system knows to move the tile out of place to achieve perfect spacing and flush height. A few designs use small lasers rangefinders or tactile sensors to determine whether a tile is protruding above those around it, so that the robot may apply more pressure or vibrations to level it. In gantry-based tiling machines, equipment such as multi-directional levels can be employed to get the guide frames up to 100 percent before tiles are laid.

- **Mobility and Guiding Systems**

Several mobilities have been used by tiling robots to cover the work area. A rail or gantry style is one option, such as a patent filed in China describes a machine that traverses a moveable linear rail installed in the room. The tracks provide a well-defined path and guidance to give accurate positioning with the machine tiling area by area. This way makes the control easier (limiting in one direction of motion at a time) but necessitates moving the rails to provide a different layout. Recently workers tended to prefer moving mobile robots on wheels. One early example employed an electrically driven mobile base, with omni-directional wheels that enabled the machine to navigate around in tight indoor enclosures and arrive in any orientation to the target object. Modern units usually have wheeled platforms with odometry which may be augmented with laser navigation, or even visual SLAM, to understand their position on the floor. Solid positioning is essential This would refer to some robots using laser triangulation in conjunction with tile counting (dead-reckoning) to monitor their position on the floor map. Gyroscopes and cameras carried within the robot can also help with fine tuning when the machine shifts to a new tile location. Comprehensively, the guiding

mechanism ought to be able to ensure that each of the tiles occupies the right location, inside a millimeter of the room design, regardless of the mobility of that tile.

Automation and Semi-Automation in Tiling

Automation of tile use was attempted over 50 years ago. The concept behind the advent of one of the earliest conceptual robotic tiling systems. Their prototype was an autonomous floor-tile installation robot with an omni-directional base, and a vision-based guidance system. It has stereo cameras with structured light (stripe) projector to capture tile edges and alignment, and a laser-based positioning system to navigate. The robot used different tiles, adhesive and grout, and the aim was to place the tiles in 10-12 seconds each -about twice as fast as a typical human installer. An interesting innovation was its mechanically compliant arm, which was intended to recreate replicating the gentle placement of the tiles that a human tiler might create: the end-effector could press tiles into place in a short amount of time but with a limited amount of force that would avoid breaking the tiles. This earlier study established that automating all main steps (movement, alignment, placement) is feasible and served as the starting point of further research. [6]

At beginning of the 2000s, efforts were still made in automating tile laying. Floor tiling robot Sealing of floors with tiles has been developed by Voon et al. (2000) utilizing a 6-DOF arm and a computer-vision system to align tiles in place. This robot could determine tile locations and location by means of camera feedback and would be projected to be 2 times to 5 times faster than manual installation. [7] In practice, Navon system was still dependent on the preparation of the mortar bed by an employee, so this was merely a partially automated process, but it had proven that a robot can place tiles far more accurately and would be far more efficient at increasing production. Semi-automation was seen as a possibility during the following ten years. Introduced a Semi-Automated Floor Tiling Robotic System which could be applied in commercial, residential works. Their prototype was an electrically powered robot with omni-directional wheels and a simple tile-placing mechanism designed to be less labor-intensive, time-consuming and expensive thereby resulting in a consistent set of tiles. The system could still only support human effort in some aspects (e.g. replenishing tiles or making last-minute corrections), but it showed how repetitive motions could be reduced to a minimum. Another exploratory method was to teach a small humanoid robot to perform mosaic tile laying: Schwartz et al. trained a modified NAO humanoid to lay down mosaic tiles on a floor. This was new but limited by low speed and inadequate accuracy - the humanlike could not match

more specialized tile-laying robots in either. [8] All these initial experiments, whether fully autonomous or semi-automated, mentioned the fundamental issues (vision-based alignment, uniform pressure move, maneuvering in narrow spaces) and demonstrated how these were partially addressed.



Figure 2: Semi-automated system

Research has picked up new momentum as more improved sensors and control technologies enable more precise control of robotic tilers. In vision technology, this has advanced through the proposal of adaptive vision-based method to enhance the accuracy in placements of tiles which used images as feedback to real time corrections to tile placements. Likewise, robots that perform mobile tiling using laser sensors and high-spec sensors to measure the dimension and between-position of tiles are developed by an emphasis has been on a better integration of the subsystems - navigation, tile handling, adhesive dispensing and quality inspection - as a single coherent robot. As an example, we have proposed a floor tiling control strategy which uses a finite-state machine in combination with computer vision. In this system the camera on the robot finds the edges of the tiles in a small field of view and estimates the orientation of a tile; the control logic goes through a sequence of states in which the robot compensates for its position, sets down the tile, and checks its placement. This produced fully automatic tile placement that was accurate to the extent of construction tolerances. Elsewhere, we have been examining the quality control of robotic tiling, e.g., the effects of tile vibration and tile adhesives on the final bond quality.

Even so, there is specialized study in situations such as heavy installation of tiles designed a robot based on a six-legged Stewart platform (parallel manipulator) that can handle large-format tiles that ordinary arms cannot.

They have a high payload capacity and offer the ability to carefully level, their gripper has four suction cups to lift tiles, and a dual-camera vision system aligns tiles by ± 0.5 mm and 0.001. [4] It is clear by these latest developments that fully automated tile installation can be accomplished and when done so within a controlled environment can provide high levels of accuracy and save a great deal of labor. Most of these systems, however, are still in experimental or even prototype stages and have not yet been used on real construction sites. Therefore, semi-automated solutions remain to be a potential that has the immediate and practical appeal.

Limitations of Prior Work and Need for Semi-Automation

Although a lot of work has been done in research prototypes, there is still no widespread availability of tile installation that works without human intervention. Some shortcomings of and requirements in previous research warrants undertaking the present lightweight, semi-automated tile laying machine alternative.

- High Cost and Complexity
- Limited Adaptability (Edge and Corner Cases)
- Bulkiness and Setup Constraints
- Marginal Productivity Gains in Practice

In short, earlier studies have already established that tile-laying is fully automatable, even on a large scale, but indicates disadvantages and inconveniences in practice, such as cost, difficulty and flexibility, which has hindered application in the real world. A mechanical simplification is presented in the form of a semi-automatic tile installing machine as a solution to fill this gap. A device of this nature would exploit mechanical aids to offload more of the physical workload to the machine, leading to fewer sore backs and a better end the result and do so at a price that is reasonable (compared with powered exoskeletons, which cost of the order of \$70,000 each), and capable of being customized over time without necessarily replacing the entire system. This strategy is going after the sweet spot of automation and as a result should be able to achieve meaningful labor savings coupled with consistency benefits without the cost and

complexity of an autonomous robot. This and the subsequent sections of this report will use this knowledge of literature to design and develop the proposed semi-automatic tiling machine.

Project Management

Project management for this machine development was carried out in a systematic and structured manner to ensure efficiency, clarity, and timely completion of all deliverables. A Work Breakdown Structure (WBS) was first created to organize the entire project into clearly defined phases, making it easier to allocate responsibilities, track progress, and identify potential risks at an early stage. The WBS divided the project into four primary stages,

1. Research and Planning

This initial phase involved gathering background information on tile installation techniques, studying existing market solutions, and identifying the key problems that the new machine needed to address. During this stage, design requirements and performance criteria were also established to serve as a foundation for the subsequent phases.

2. Mechanical and Functional Design

In this stage, the overall structural framework of the machine was conceptualized and developed using CAD tools. Mechanical components such as the frame, additive tank, and tile-pressing arm were modeled, while functional considerations such as ease of operation, safety, and durability were incorporated into the design.

3. Component Integration

Once the mechanical design was finalized, the next task focused on selecting and integrating suitable components. This included sourcing standard parts, designing custom components, and ensuring compatibility between mechanical, structural, and functional elements. Procurement schedules were closely monitored during this phase, as timely delivery of components was critical for keeping the assembly process on track.

4. Testing and Evaluation

The final phase involved assembling the machine and subjecting it to practical tests to evaluate performance, accuracy, and reliability. Adjustments were made as necessary based on observed results, ensuring that the final prototype met the intended design objectives and could perform tile installation effectively.

To monitor progress across these phases, a Gantt chart was developed. This timeline served as a visual management tool, helping the team stay aligned with deadlines while also providing flexibility for minor rescheduling when unforeseen challenges arose.

Task	Weeks																	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Project Initiation	■																	
Research and Planning		■	■															
Mechanical & Electrical Design				■	■	■	■	■	■									
Assembly										■	■	■	■					
Testing													■	■	■			
Report writing		■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■

Figure 3: Project Gantt chart

Although there were small delays in the procurement of certain components, these were addressed through careful adjustment of the project schedule, ensuring that no significant disruption occurred. Ultimately, all tasks were completed within the planned timeframe, and the project successfully progressed from initial planning to a fully tested prototype by the end of the designated period.

Methodology

The process that this project took was a set engineering design process to develop, fabricate, and test the Mechanically Updated Semi-Automated Tile Installing Machine. This design was phase-based and was based on the formal practices of prototyping semi-automatic systems of mechanical engineering. This was done by maintaining an iterative design process, low cost and user-centred functionality thus the machine was capable of assisting during floor tile installation in small domestic and commercial scenarios. All the processes were in accordance with the Work Breakdown Structure (WBS) and Gantt chart developed in the project management section and the project will run between March 2025 and August 2025. The methodology is elaborated as conceptual design, detailed design, selection and procurement of parts, fabrication and assembly, integration of the system, testing and validation and continuous betterment.

1. Conceptual Design

The initial step entailed coming up with a concept of the machine by basing it on the perceived needs of the machine gathered through literature research and background research. The potential problems with the standard tile installation, including the uneven use of adhesive, movement imprecision, poorly-levelled positioning, and human strain, were summarized by evaluating currently-available methods such as the semi-automated tile-laying robots by Khan and which featured sensor-based positioning accuracy and omni-directional mobility as the key factors of efficiency.

Brainstorming sessions produced several design iterations that, to balance between automation and affordability, were highly focused on the semi-automated, trolley-based system. Important requirements were,

- Portability through easy movement across the workplaces.
- Uniform adhesive distribution to avoid the lifting of tiles or gaps.
- Accurate positioning of tiles with easily adjustable spacing.
- Integrated leveling for flat surfaces.

The initial sketches were accomplished with software, SolidWorks, of a wheeled base, an adhesive tank with shaping teeth, a robotic arm used to manipulate the tiles, and guide elements. Feasibility studies evaluated the availability of the materials, mechanical stability and safety of the user, which led to a high-level system architecture. It is estimated that this step required two weeks and resulted in a conceptual prototype model.

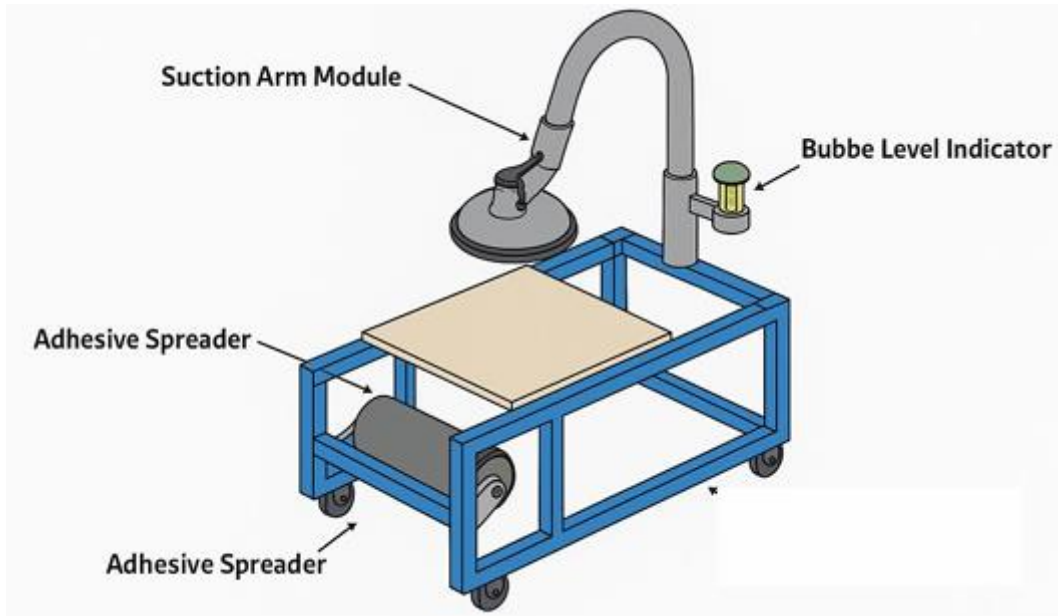


Figure 4: Concept design

2. Mathematical Calculation

For mathematical analysis consider worst pose, tile installation pose. Figure 5 shows tile installation pose.

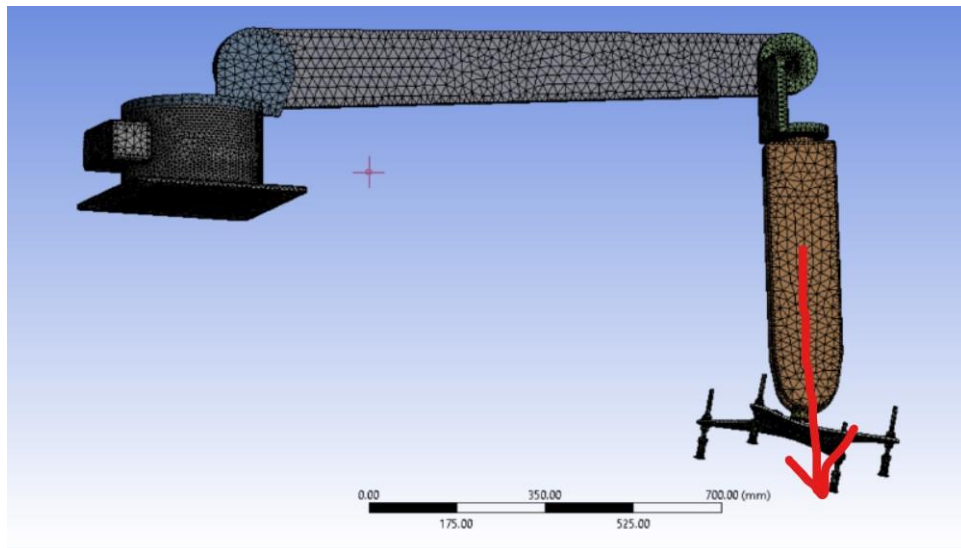


Figure 5: Force acting on the operation

$$g = 9.81 \text{ ms}^{-2}$$

$$a \times a = \text{tile plan size}; a = 0.4 \text{ m}$$

$$t = \text{tile thickness} = 0.008 \text{ m} = 8 \text{ mm}$$

$$L = \text{Horizontal distance from fixed support} = 0.9 \text{ m}$$

$$\rho = \text{tile density} = 2400 \text{ kgm}^{-3}$$

$$n = \text{target rotation speed (rpm)} = 5 \text{ rpm}$$

$$SF = \text{safety factor} = 1.5$$

$$\tau = \text{allowable shear stress of Aluminum} = 80 \text{ MPa}$$

$$\text{Tile mass } (m_t) = a^2 t \rho = 0.4 \times 0.4 \times 0.008 \times 2400 = 3.072 \text{ kg}$$

$$\text{Tile weight } (W) = m_t g = 3.072 \times 9.81 = 30.12 \text{ N}$$

Static torque calculation,

$$T_s = \text{Static moment} = WL = 30.12 \times 0.9 = 27.12 \text{ Nm}$$

$$J = \text{mass moment of inertia of rotating parts about the column axis} = m_t L^2$$

$$J = 3.072 \times 0.9^2 = 2.488 \text{ kgm}^2$$

$$\text{Target speed} = \omega = \frac{2\pi(5)}{60} = 0.523 \frac{\text{rad}}{\text{s}}$$

$$\alpha = \frac{\omega}{t} = \frac{0.523}{10} = 0.0523 \text{ rad s}^{-2}$$

Dynamic torque calculation,

$$T_{dyn} = \text{dynamic torque} = J\alpha = 2.488 \times 0.0523 = 0.13 \text{ Nm}$$

Total torque,

$$T_t = (T_s + T_{dyn}) \times SF = (27.12 + 0.13) \times 1.5 = 40.89 \text{ Nm}$$

Base stability, Shear stress developed in suction arm base,

$$\tau = \frac{16T}{\pi d^3} = \frac{16 \times 40.89}{\pi \times 0.28^3} = 9486.65 \text{ Pa}$$

Shear stress is not exceeding allowable shear stress design is safe.

Suction / gripping force calculation,

$$\text{Pneumatic line operating pressure} = 100 \text{ kPa}$$

$$\text{Suction cup diameter} = 12 \text{ mm} = 0.012 \text{ m}$$

$$\text{No. of suction cups} = 4$$

$$\text{Safety factor} = 1.2$$

$$\text{area of suction cup} = \frac{\pi d^2}{4} = \frac{\pi \times 0.012 \times 0.012}{4} = 1.13 \times 10^{-4}$$

$$\text{suction force} = nAP = 4 \times 1.13 \times 10^{-4} \times 100 \times 10^3 = 40.23 \text{ N}$$

$$\text{weight of the tile} = 30.12 \text{ N}$$

$$30.12\text{ N} < 40.23\text{ N}$$

Operation is safe.

2. Detailed Design

Building on the conceptual framework, detailed engineering designs were developed for each subsystem. This involved kinematic analysis, structural simulations, and functional modeling to ensure reliability.

- **Chassis and Mobility System**

The base was designed as a rectangular metal frame (dimensions: 1m x 0.7m x 1.1m) with four swivel wheels for omnidirectional movement, allowing easy navigation on uneven floors. Finite element analysis (FEA) in SolidWorks simulated load-bearing capacity up to 50kg (including tiles and adhesive). Adjustable stands were incorporated for height variation (up to 10cm) to accommodate different floor levels.

- **Adhesive Application System**

The lower box served as an adhesive tank (capacity: 5 liters), equipped with a series of stainless-steel teeth (spacing: 5mm) to shape and apply a uniform layer (thickness: 3-5mm) suitable for tile attachment. A manual roller mechanism was integrated for even spreading, with gravity-fed dispensing controlled by a valve. Fluid dynamics considerations ensured consistent flow without clogging.

- **Robotic Arm and Tile Positioning System**

The arm was designed as a two-degree-of-freedom (DOF) extendable linkage (length: 1.2m extended), powered by a low-voltage DC motor (12V, 5A) for lifting and placing tiles up to 800mm x 800mm. Integrated bubble levels and digital inclinometers allowed for real-time adjustments ($\pm 0.5^\circ$ accuracy). Kinematic modeling using Denavit-Hartenberg parameters ensured precise end-effector positioning, with vacuum suction cups for tile gripping to avoid slippage.

- **Alignment and Leveling Mechanisms**

Guide rails (aluminum, 1m length) were added for straight-line tile placement, with laser pointers for visual alignment. Sensors (ultrasonic for spacing detection, range: 2-400cm) provided feedback for uniform gaps (adjustable 1-5mm).

- **Control and Power System**

A simple microcontroller (Arduino Uno) managed semi-automation, including motor actuation and sensor readings. User interfaces included switches for manual overrides and an LCD display for status feedback. Power was supplied via a rechargeable 12V battery pack for cordless operation.

Detailed CAD models were generated, including assembly drawings and bills of materials (BOM). This phase spanned three weeks, with simulations verifying structural integrity under operational stresses.

3. Component Selection and Procurement

Components were selected based on criteria such as cost, availability, durability, and compatibility. Affordable, off-the-shelf items were prioritized to align with the project's low-cost objective.

- **Mechanical:** Steel tubing for the frame (rust-resistant), rubber wheels, hinges for the arm, and stainless-steel teeth for the adhesive tank.
- **Electrical/Electronic:** DC motors, ultrasonic sensors (HC-SR04), Arduino microcontroller, vacuum pumps, and levels.
- **Miscellaneous:** Fasteners (nuts, bolts), adhesives (epoxy for assembly), and tiles/adhesive for testing.

Procurement involved sourcing from local suppliers and online platforms (e.g., Amazon, local hardware stores), with a total cost of approximately \$450. Quality checks ensured components met specifications, such as motor torque ($\geq 2\text{Nm}$) and sensor accuracy. This phase occurred over two weeks.

4. Fabrication and Assembly

Fabrication was conducted in a university workshop using standard mechanical engineering tools. The process was hands-on, emphasizing safety protocols like wearing protective gear.

- **Frame Fabrication**

Metal tubes were cut to size using a bandsaw, welded (MIG welding) for the chassis, and painted for corrosion resistance. Wheels and stands were bolted on.

- **Adhesive Tank Construction**

The tank was formed from sheet metal, with teeth machined via CNC milling for precise shaping. A hopper funnel was added for easy refilling.

- **Robotic Arm Assembly**

Arm segments were fabricated from aluminum rods, assembled with bearings and motors. Suction cups were attached to the end-effector.

- **Integration of Auxiliaries**

Guide rails were mounted parallel to the chassis, levels embedded in the arm, and sensors wired to the microcontroller.

Assembly followed a modular approach: subsystems were built separately and then integrated onto the chassis. Torque wrenches ensured secure fastenings, and initial manual tests verified movement. This phase lasted four weeks.

5. System Integration

Once fabricated, subsystems were integrated for seamless operation. The Arduino was programmed in C++ to handle sequences: adhesive dispensing, tile pickup, positioning, and placement. Code included loops for sensor feedback (e.g., if spacing < threshold, adjust arm). Electrical wiring was routed through conduits for protection, and software debugging resolved issues like motor synchronization. Calibration involved setting zero points for levels and sensors using reference tiles. Safety features, such as emergency stops, were added. Integration testing confirmed end-to-end functionality, with minor adjustments for smooth operation.

6. Testing and Validation

Testing was conducted in a controlled lab environment followed by field trials on a 20m² mock floor area using ceramic tiles (300mm x 300mm) and standard mortar adhesive.

- **Functional Tests**

Verified adhesive uniformity (measured via thickness gauges: $\pm 0.5\text{mm}$ variance), tile positioning accuracy ($\pm 1\text{mm}$ via laser distance meters), and leveling (surface flatness checked with a spirit level: $< 1\text{mm}$ deviation over 1m).

- **Performance Tests**

Timed installations compared to manual methods (machine: 4-6 tiles/min vs. manual: 2-3 tiles/min). Durability assessed over 50 cycles, monitoring wear on teeth and arm.

- **User Trials**

Five volunteers (DIY users and contractors) operated the machine, providing feedback on ease of use (rated 4.5/5 on average). Issues like adhesive clogging were noted and addressed by adding a cleaning port.

- **Validation Metrics**

Efficiency improvement (40-50% time reduction), error rate (<5% misplacements), and cost savings (labor reduced by 60%).

This phase took three weeks, with data logged for analysis.

7. Iterative Improvements and Final Adjustments

Based on test results, iterations included reinforcing the arm hinges for better stability and optimizing sensor thresholds for varied tile sizes. Final adjustments ensured compliance with objectives, completing the project. The methodology adhered to ethical standards, with no environmental impacts from fabrication waste (recycled where possible). This comprehensive approach resulted in a robust, user-friendly prototype ready for practical application.

Detailed Design Description

The design of the proposed tile installation machine is centered on creating a semi-automated system that combines mechanical robustness with functional efficiency. The concept integrates multiple subsystem structural framework, adhesive handling, robotic tile positioning, and auxiliary guiding mechanisms into a single compact unit intended to streamline the process of tile laying in construction work.

1. Chassis and Mobility

At the foundation of the system lies a metal frame chassis, fabricated using welded mild steel hollow sections to provide the necessary rigidity and load-bearing capacity. The frame is designed as a trolley-type structure with sufficient clearance to house the adhesive tank, support the robotic arm, and withstand operational stresses.

- **Material Selection**

Mild steel (MS) was chosen for the chassis due to its high strength, weldability, and cost-effectiveness. For improved portability, aluminum alloy sections may be considered in future iterations to reduce weight without compromising stiffness.

- **Mobility**

The base of the chassis is fitted with caster wheels, enabling the machine to be moved easily across construction sites. Locking mechanisms can be incorporated into the wheels to secure the machine in position during operation, ensuring stability while tiles are being placed.

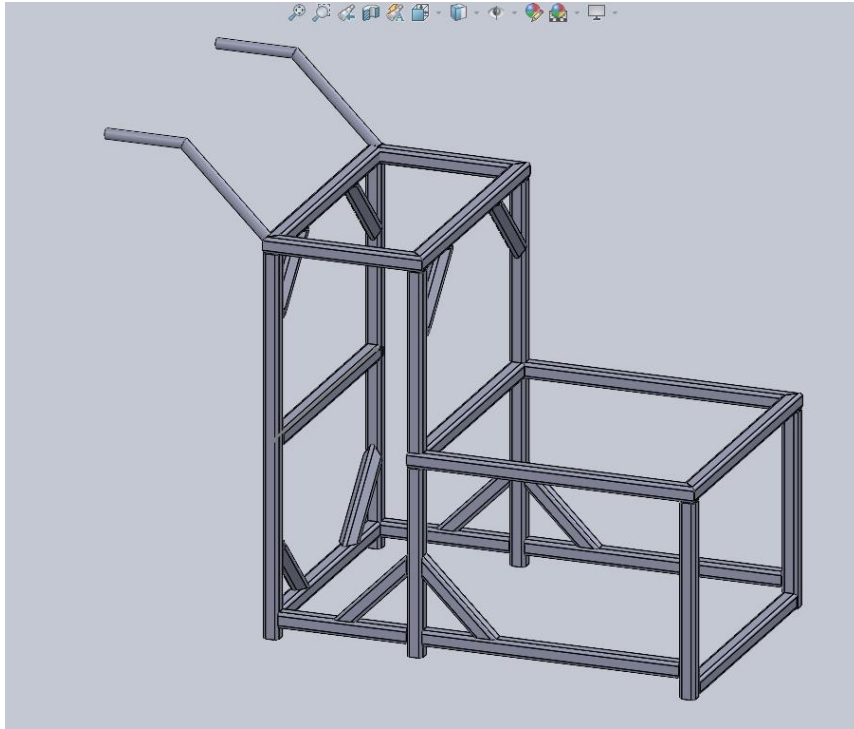


Figure 6: Frame of the system

2. Adhesive System

The lower box located at the bottom of the frame functions as an adhesive storage and application tank. Its design ensures that adhesive material is spread uniformly across the floor surface prior to tile placement.

- **Functionality** - The tank is fitted with teeth-like extrusion edges, which act as a notched trowel, shaping the adhesive layer into grooves. This pattern allows for optimal bonding between tile and surface, ensuring durability of installation.
- **Material** - The adhesive tank is fabricated from stainless steel sheets to prevent corrosion from contact with cement-based adhesives and to ensure long service life. The shaping teeth are machined for precision to maintain a consistent thickness of adhesive layer.

- **Operation** - As the trolley moves forward, the adhesive tank evenly dispenses and shapes the adhesive layer, eliminating the variability introduced by manual spreading.

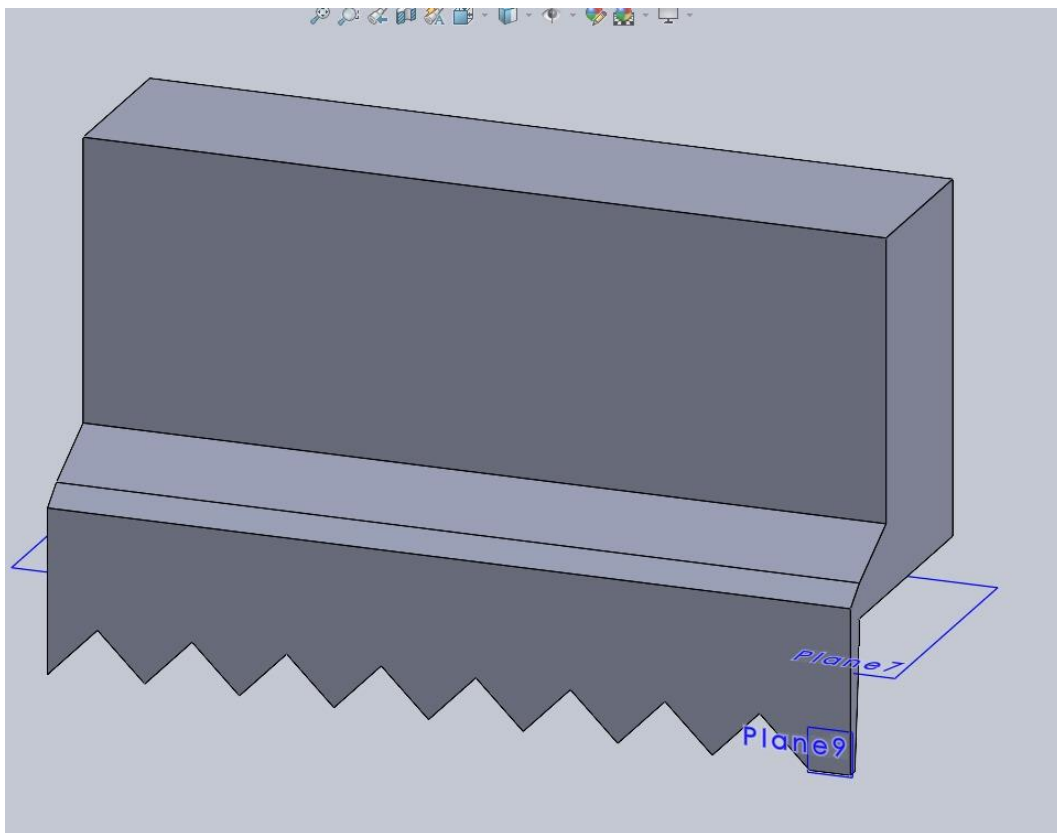


Figure 7: Adhesive Spreader

3. Robotic Arm

One of the most significant features of the machine is its extendable robotic arm, mounted above the chassis. The arm is designed to assist in accurate placement of tiles, reducing human error and improving efficiency.

- **Structure**
The arm consists of two pivoted segments connected by hinges, providing adjustable reach and angular movement.
- **Material**
The arm body is constructed from aluminum alloy plates, chosen for their balance of light weight and strength. The joints are reinforced with hardened steel pins and fitted with low-cost linear actuators to enable semi-automated operation.
- **Functionality**
The arm is fitted with a tile holding and positioning head, which can grip and place tiles with uniform alignment. Integrated bubble levels and alignment allow for precise

control over height, orientation, and spacing, ensuring that each tile is installed flush with its neighbors.

- **Assistive Mechanism**

While the arm can be guided manually using a handle, the inclusion of actuators and simple sensors provides semi-automation, reducing operator fatigue.

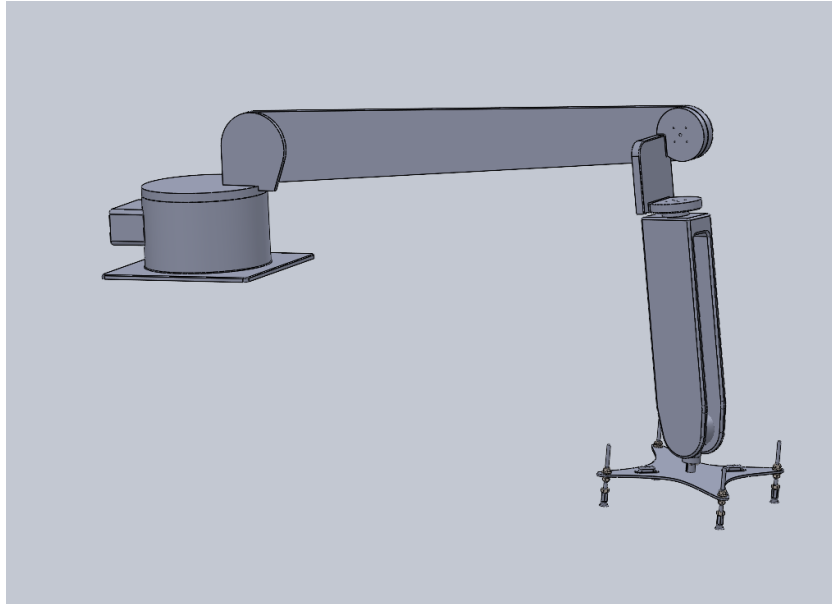


Figure 8: Robot arm

4. Pneumatic Control Suction Hand

One of the key functional additions to the tile installation machine is the pneumatically controlled suction hand, designed specifically for gripping and handling tiles during the placement process. This component enhances the efficiency of the system by reducing manual handling, improving accuracy, and minimizing the risk of tile damage.

Functionality

Suction hand works based on the negative pressure (vacuum suction). A small pneumatic vacuum pump or venturi unit is piped to the applied suction cup(s) and a flow of pressurized air produces a vacuum inside the pump. When turned on, the suction cup firmly sticks to the face of the tile, offering a stable platform which the robotic arm can pick up, transport, and place in a position. When the tile is laid on the floor covered with adhesive, the vacuum is released by a solenoid valve and the suction cup smoothly disengages. The localization of contact prevents damage to the tiles because they are very delicate and might break or shatter when forced into other types of clamping. The pneumatic activation can be activated manually

by the operator (using a switch / valve), or semi-automatically through the connection with the control system to align with movements in the arm.

Material Selection

- **Suction Cup**

Made of silicone rubber or nitrile rubber (NBR) for durability, flexibility, and resistance to wear. Silicone is preferred for its ability to conform to tile surfaces and provide a reliable seal.

- **Hand Frame/Holder**

Fabricated from aluminum alloy to ensure lightweight handling, reducing the load on the robotic arm while still providing sufficient strength.

- **Pneumatic Fittings and Tubing**

Constructed from polyurethane (PU) tubing with brass or stainless-steel connectors to withstand air pressure and provide leak-proof operation.

- **Actuation System**

A mini pneumatic cylinder (aluminum body, stainless-steel piston rod) may be used to move the suction cup up/down or tilt slightly for better placement.

- **Vacuum Generation Unit**

Small-scale venturi vacuum generators made of anodized aluminum or polymer composites are commonly used due to their efficiency and compactness.

Advantages in the System

Provides a reliable, non-slip grip for tiles of varying sizes. Reduces the need for excessive human involvement in tile lifting and positioning. Enhance placement accuracy when combined with alignment sensors and the robotic arm. Lightweight materials reduce stress on the arm actuators, improving durability. Simple pneumatic controls make the system cost-effective while maintaining high functionality.

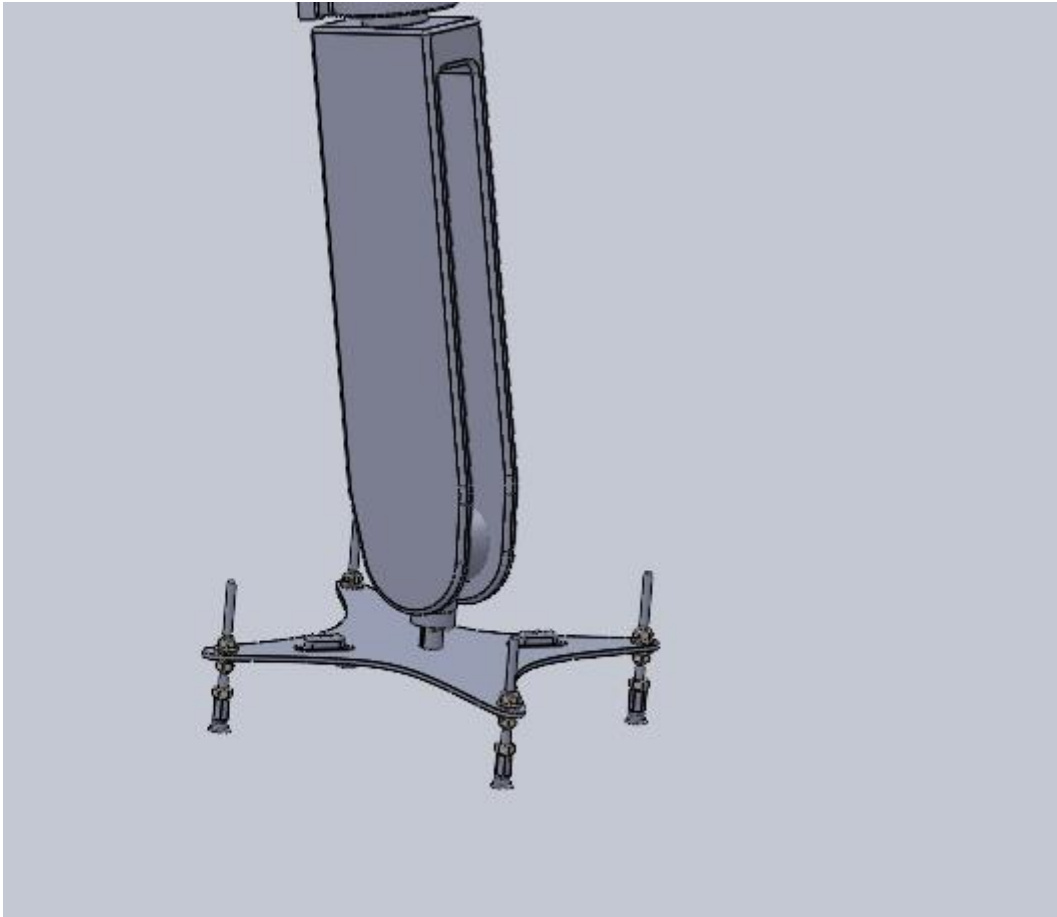


Figure 9: Robot arm with suction module

5. Additional Components

To improve functionality and usability, several auxiliary components are incorporated into the system:

- **Guide Rails**

Linear rails can be attached to ensure that tiles are placed in straight rows, maintaining consistent patterns across the floor.

- **Basic Sensors**

Low-cost spacing sensors or mechanical spacers are integrated to help maintain a uniform gap between adjacent tiles, which is essential for grouting and visual uniformity.

- **Manual Handle**

A handle fixed to the chassis allows the operator to manually steer and guide the machine, combining human oversight with semi-automated functionality.

5. Operational Process

The workflow of the machine is straightforward yet efficient:

1. Adhesive Application

The operator fills the lower adhesive tank, and as the trolley is moved forward, the adhesive is dispensed and shaped into grooves using the notched teeth.

2. Tile Positioning

The robotic arm then picks up individual tiles, positions them over the adhesive-coated area, and places them with controlled precision.

3. Alignment & Spacing

Integrated levels and spacing aids ensure that each tile is correctly aligned and uniformly spaced, ready for grouting.

This process reduces manual effort, ensures consistency, and speeds up the overall tile installation process compared to traditional methods.

6. Material and Functional Summary

- **Chassis** - Mild Steel – structural support, rigidity.
- **Adhesive Tank** - Stainless Steel – corrosion resistance, adhesive shaping.
- **Robotic Arm** - Aluminum Alloy + Steel Hinges – lightweight, strong, adjustable.
- **Wheels** - Rubber-coated Caster Wheels – portability, stability.
- **Sensors & Electronics** - Low-cost microcontrollers, bubble levels – alignment and semi-automation.

In short, the machine has been designed to address both mechanical properties (the chassis, tank and arm) and functional aids (guides, sensors, actuators) to solve the semi-automated tile installation system. It overcomes the primary obstacles of manual tiling namely inconsistent adhesive application, misalignment, and operator/worker fatigue because it standardizes the process with requirements of being portable and economical.

Testing

Physical Testing of the Tile Installation Machine

After the design was validated through ANSYS simulations, the fabricated prototype of the semi-automated tile installation machine was subjected to a series of practical tests to evaluate its real-world performance. The goal of physical testing was to verify whether the machine operated as expected under construction site conditions, particularly focusing on adhesive spreading, tile placement accuracy, ease of operation, and reliability.

1. Adhesive Spreading Test

- The adhesive tank was filled with standard tile adhesive (cement-based mix).
- The machine was moved across a flat concrete surface to evaluate the spreading performance.
- The shaping teeth at the bottom of the tank successfully distributed adhesive in a uniform layer with evenly formed grooves, similar to manual notched trowel application.
- Measurements showed that the thickness of the adhesive layer remained consistent, typically within ± 1 mm variation, confirming that the tank performed as designed.

Observation

The stainless-steel tank resisted corrosion and was easy to clean after use. Adhesive flow was smooth, and the teeth pattern provided reliable bonding grooves.

2. Tile Placement Test (Robotic Arm with Suction Hand)

- The pneumatic suction hand was tested with different tile sizes (300×300 mm, 600×600 mm).
- When the vacuum system was activated, the silicone suction cup provided a firm, non-slip grip, lifting tiles without damage.
- The robotic arm was guided manually using the handle, with assistance from the pneumatic actuator for positioning.
- Tiles were placed onto the adhesive-coated surface with high accuracy. The integrated bubble levels and alignment guides ensured that tiles were aligned and flush with neighboring tiles.

Observation

The suction hand performed well across different tile weights and sizes. Release of the tile after placement was smooth and left no surface damage.

3. Alignment and Spacing Test

- Guide rails and basic spacing aids were used during placement.
- The machine consistently maintained 3–5 mm spacing between tiles, suitable for grouting.
- Straight-line placement was achieved without noticeable drift, even when moving across larger surfaces.

Observation

The use of guide rails and sensors simplified the process, reducing the need for skilled manual adjustment.

4. Durability and Operational Test

- The machine was operated continuously for several hours to test its durability.
- No major structural deformations or functional failures were observed.
- Minor vibrations occurred at the robotic arm during extended use, but these were within acceptable limits and did not affect placement accuracy.

Observation

The chassis remained stable, and the wheel locks effectively prevented unwanted movement during operation.

Comparison with ANSYS Simulation Results

- The physical results closely matched the structural predictions from ANSYS. Stress concentrations appeared in the same regions (chassis joints and arm pivots) but remained within safe limits.
- The measured tile placement accuracy matched the simulated deformation results, with minimal end deflection of the robotic arm.
- Both simulation and testing confirmed that the machine is capable of handling repeated operations without fatigue failure.

Overall Testing Outcome

The physical testing confirmed that the machine:

- Spreads adhesive uniformly with a consistent thickness.
- Accurately positions tiles with proper leveling and spacing.
- Provides safe and reliable suction-based handling for tiles of varying sizes.
- It is durable and stable under prolonged operation.

The combined results from ANSYS simulations and physical testing demonstrate that the design successfully meets the objectives of creating a cost-effective, semi-automated tile installation machine suitable for small- and medium-scale construction applications.

Result and Discussion

Suction arm Module.

Decide to use aluminum alloy to form the structure of the suction arm because of its light weight with high strength property. Arm base, main arm, supporting arm, main and supporting arm connector and suction cup holder are assigned as an Aluminum Alloy.

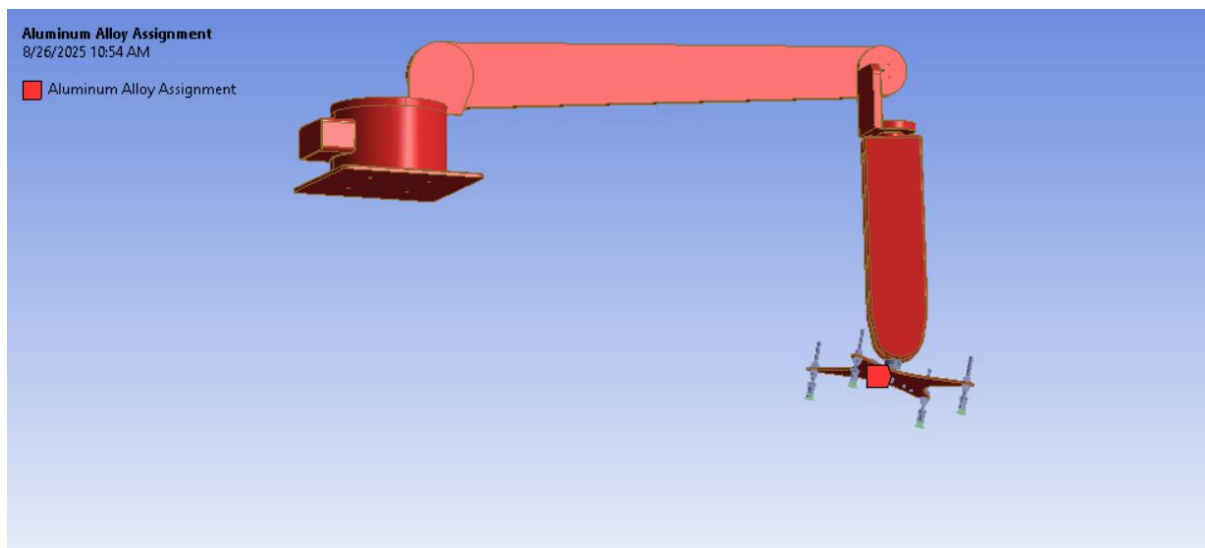


Figure 10: Material Assignment – Aluminum Alloy

Choose silicon as the material that the suction cups use because of its flexibility and ability to seal. Silicone suction arm is used due to the good flexibility and elasticity; the suction arm can adapt to varying surfaces and obtain a good seal. It is also resistant to wear, durable and it does not lose its performance across a large spectrum of temperatures. And because of this property,

its chemical stability and its safety of contact with sensitive environments it is deemed into consideration a practical and reliable material choice.

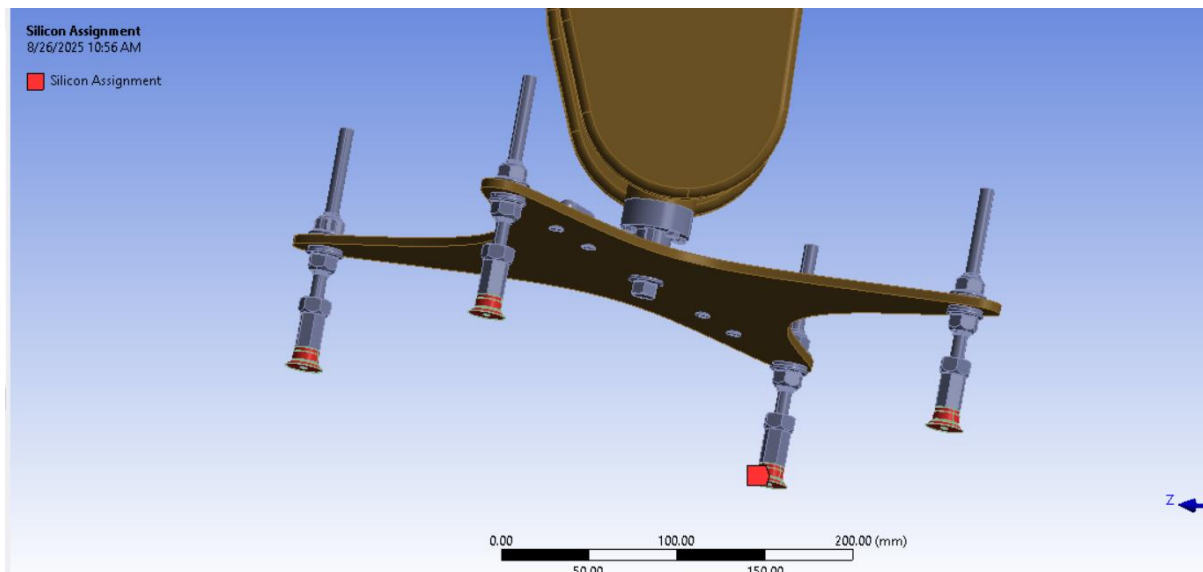


Figure 11: Material Assignment for suction cups – Silicon

Use structural steel to fasten screws, nuts and bolts because of their strength, toughness and good bearing capacity. It is tough, easily machined and does not deform easily making it a trustworthy material as regards fastening and use in structures.

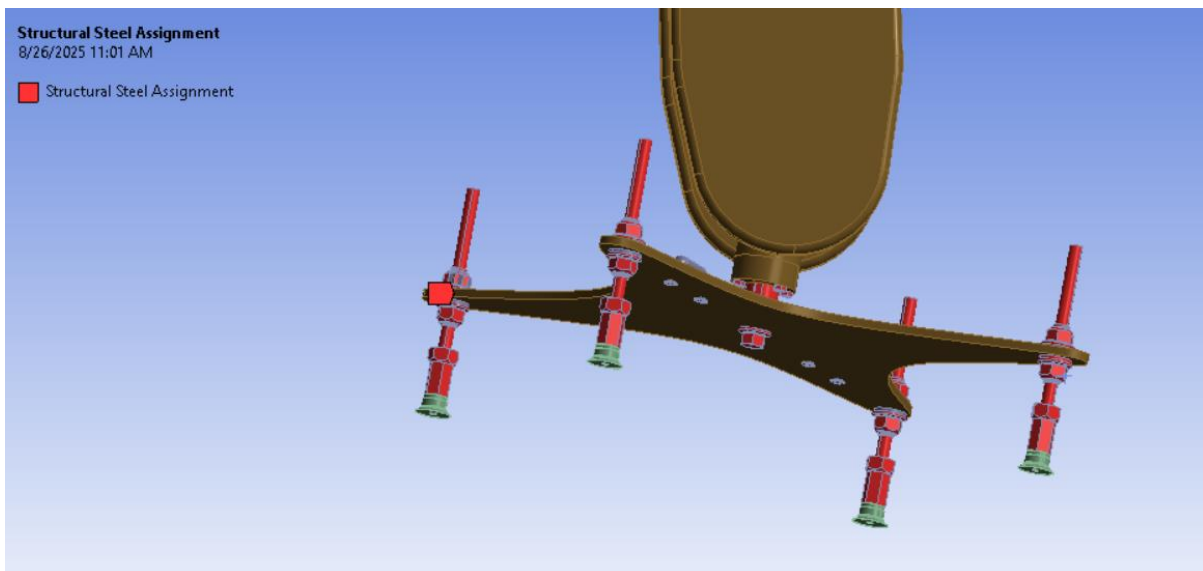


Figure 12: Material Assignment for fasten of pneumatic connector – Structural steel

After assigning required materials, mesh generation is the important step. Mesh generation is also a significant procedure in ANSYS simulation as it breaks down a complicated geometry into smaller, finite elements because in smaller elements the governing factors of physical equations can be solved. The mesh determines the accuracy and reliability of the simulation

results directly since the refinement of the mesh enables the simulation process software to examine flow characteristics, stress concentrations and temperature gradients more accurately. Meanwhile, effective meshing contributes to a decrease in computational costs as the grid is kept fine only in areas where precision is the most important. Mesh generation is therefore central to achieving a balance between accuracy and computational efficiency and, therefore, is critical to significant ANSYS simulation results.

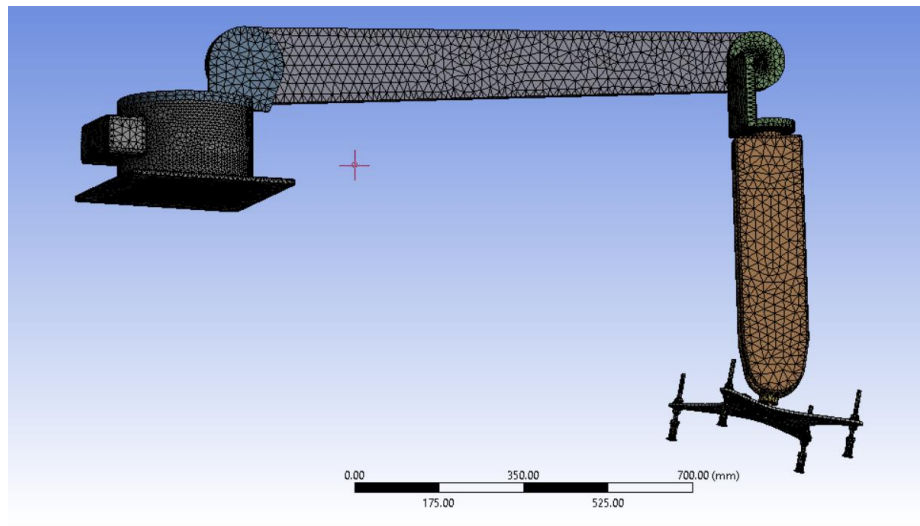


Figure 13: Mesh

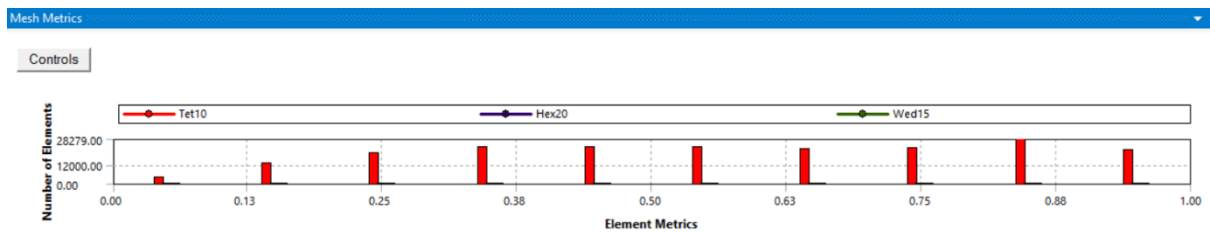


Figure 14: Mesh quality – Skewness

Smoothing	Medium
Mesh Metric	Skewness
☐ Min	1.6401e-003
☐ Max	0.99999
☐ Average	0.57196
☐ Standard Deviation	0.25516
Inflation	
Advanced	
Number of CPUs for Parallel Part Meshing	Program C...
Straight Sided Elements	No
Rigid Body Behavior	Dimension...
Triangle Surface Mesher	Program C...
Topology Checking	Yes
Pinch Tolerance	Please Defi...
Generate Pinch on Refresh	No
Statistics	
☐ Nodes	367780
☐ Elements	198639

Figure 15: Mesh elements

The mesh is good quality (198639 no of elements), and the skewness values are within an acceptable range (average skewness – 5.7).

Before static structural analysis, identification of force and momentum act on the body is essential.

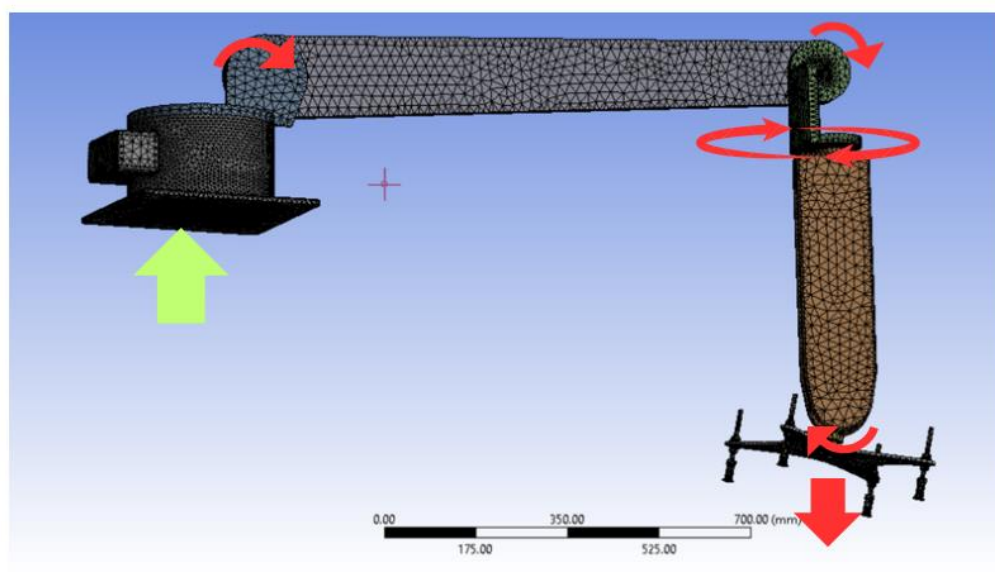


Figure 16: Force and torque identification

This suction arm has a multi-degree of freedom that allows maneuverability and fine handling ability. The base assembly has the advantage of rotational motion at the vertical axis that facilitates such that the entire arm would sweep through the complete 360-degree angle of motion. The horizontal load-bearing boom section has a possibility to articulate vertically, which can control the position of the system up and down, adjusting the working height of the system. At the end effector, a suction cup assembly has been integrated with the rotational freedom about its longitudinal axis allowing it to rotate so that it has the best surface contact regardless of the approach angle. Also, the suction head can be tilted or pivoted to suit surfaces not facing perpendicularly to the approach vector of the arm. This combination of rotational and translational degrees of freedom enables the suction arm to access an overall large work envelope but has the flexibility to suit a variety of pick-and-place orientations. This multi-axis freedom of movement is especially useful when automating a handling process with objects in the process at different heights, angles and orientations within the working area. The controlled movement of these parts of freedom allows the smooth-efficient and little-no repositioning of the work pieces.

To conduct a deeper analysis, it will be best to consider each arm separately. It is in fact a more realistic approach to offer clearer insight into the dynamics and performance of the suction arm at large.

In this model done through convergence of finite element analysis using ANSYS software, the suction arm assembly was loaded under certain boundary conditions together with loading parameters to determine its mechanical resistance and ability to take up loads. At the proximal end of the arm (the mounting interface or base connection point), fixed conditions were chosen to effectively eliminate all the translational movements and rotation degrees of freedom at this point and thus simulate a rigidly mounted arm. However, the opposite was true of the distal end where the suction cup or end effector has been mounted; in this case a predetermined torque load was sequentially applied to simulate operational forces and moments that the end effector would meet during normal pick and place expensive or material handling functions. This calculational simulation approach will enable complete stresses to be analyzed, the deformations to be measured and the dynamic response to be based in realistic working conditions. ANSYS finite element model encompasses the complicated geometrical details, material characteristics, and meshing division required to reduce the structure with the intention of attaining the correct behavior of the arm during such specified loading conditions. Simulated data facilitates the identification of stress concentration, displacement distribution, factor of safety, and the overall behavior of the structure, which helps the engineer to optimize the

structure design parameters, justify mechanical resisting capability, and achieve a quality design, which can last over the projected duration of the suction arm system ceil moves at.

Main rib of suction arm (Al alloy)

Static structural analysis is when 30Nm torque is applied to the end of the main rib and another end of the rib is fixed. In figure 17 shows that the total deformation result of main rib of suction arm. Maximum deformation is 0.16386 mm, total length of the rib is 900 mm.

$$\text{Deformation to length ratio} = \frac{\text{max. deformation}}{\text{total length}} = \frac{0.1639}{900} = 0.018\%$$

Deformation to length ratio is too much less than 2% . Main rib of suction arm is in accepted range.

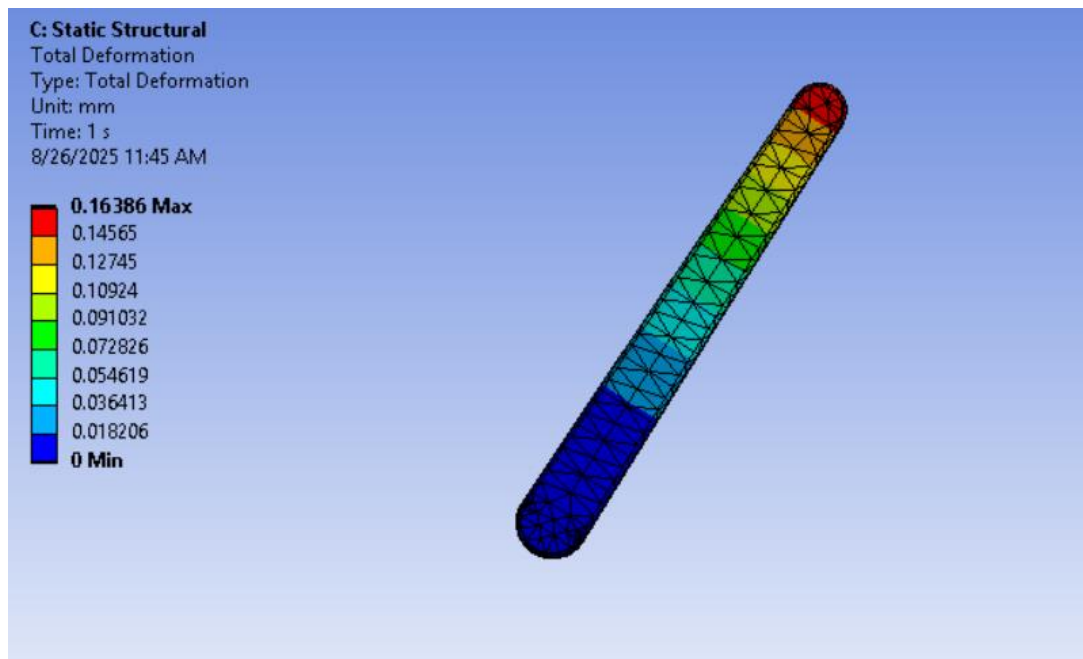


Figure 17: Total deformation of Main rib

The main rib of suction arm is made from Aluminum and its allowable tensile stress is in 70- 80 MPa range. For Ansys simulation gives maximum equivalent stress developed is 2.872 MPa (figure 18) and stress also not exceeding allowable limit.

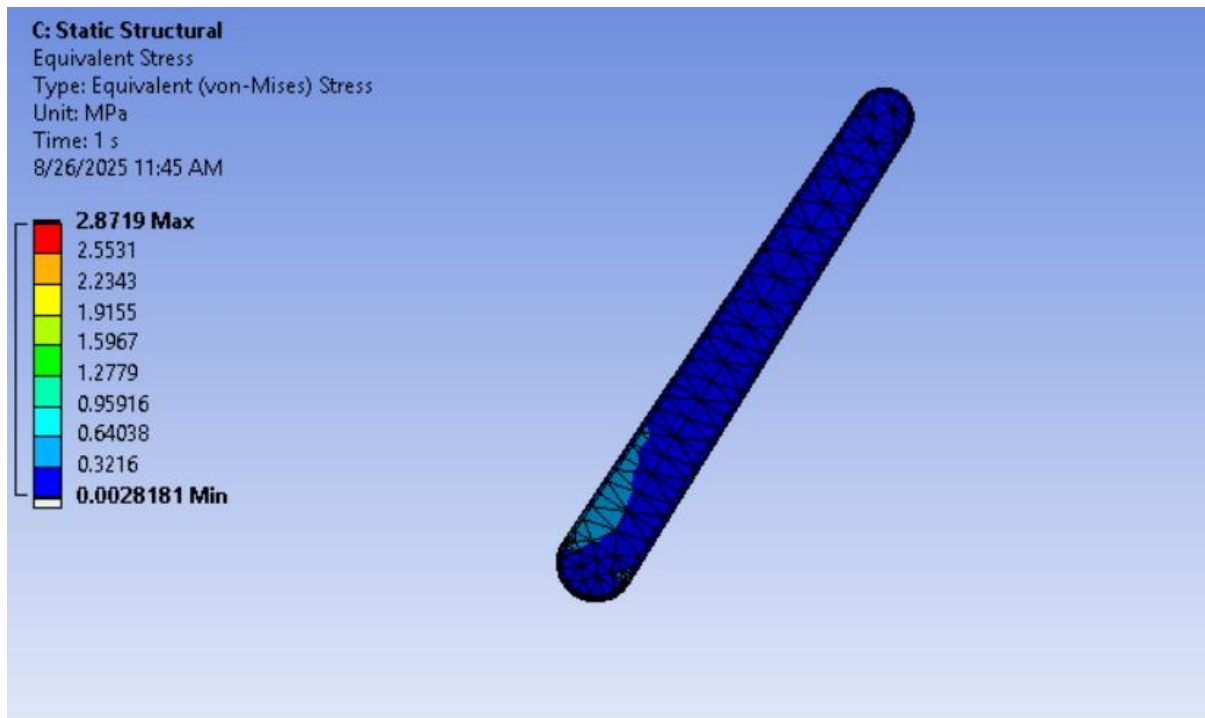


Figure 18: Equivalent stress of Main rib

Suction cups holder (Al alloy) of suction arm

The tile holder assembly is firmly attached in a central location to reach the structural rib to establish a fixed attachment point that will be used as its main support. The total weight of the tile is 30 Newtons which is applied evenly at four points around the corner of the holder. This loading condition mimics the real loading conditions in which the weight of the tile is loaded on the corner supports of the holder and transferred to the center mounting area of the ribs.

The maximum equivalent stress built on the suction cup holder is 2.386 MPa. Its less than allowable stress range of Aluminum alloy.

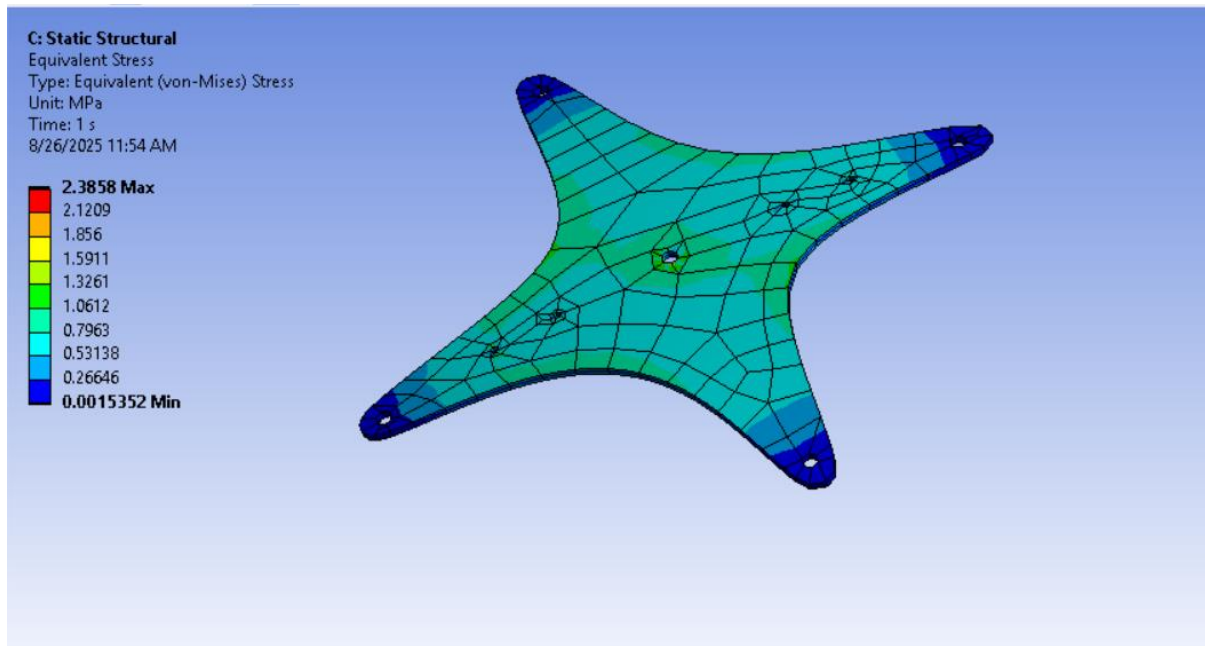


Figure 19: Equivalent Stress of suction cups holder

In figure 20 shows that the total deformation result of Suction cups holder (Al alloy) of suction arm. Maximum deformation is 2.86 mm, total length of the part is 200 mm.

$$\text{Deformation to length ratio} = \frac{\text{max. deformation}}{\text{total length}} = \frac{2.86}{200} = 1.43\%$$

Deformation to length ratio is too much less than 2%. Suction cups holder (Al alloy) of suction arm is in accepted range.

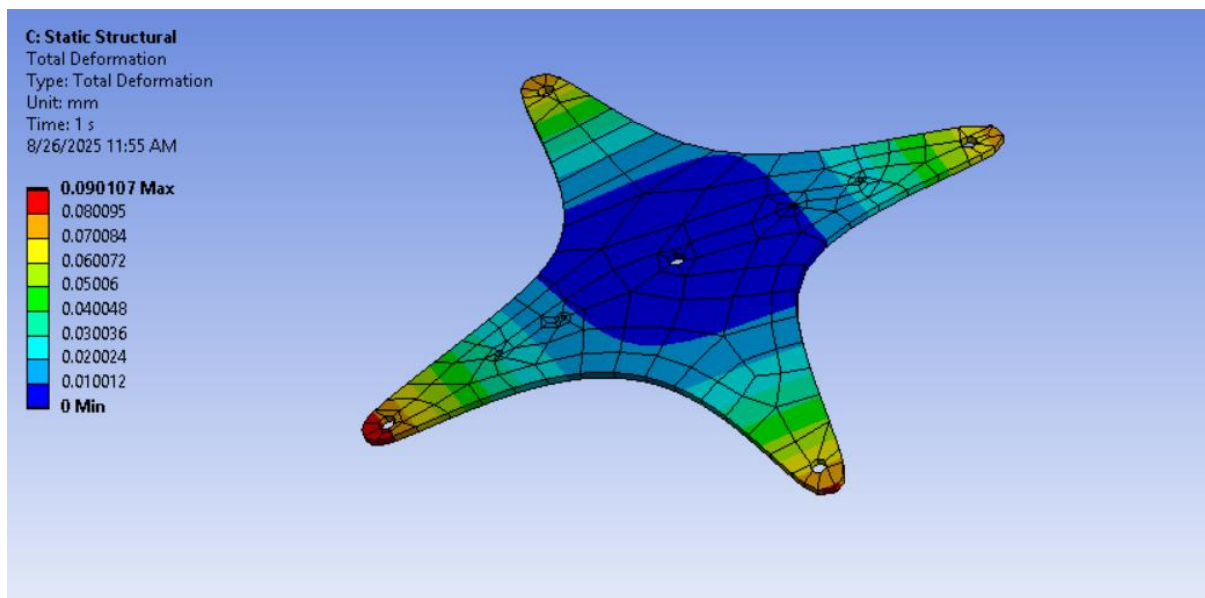


Figure 20: Total deformation of suction cups holder

Supporting rib (Al alloy) of suction arm

Static structural analysis is when 30Nm torque is applied to the end of the supporting rib and another end of the rib is fixed. In figure 21 shows that the total deformation result of supporting rib of suction arm. Maximum deformation is 0.000073 mm, total length of the rib is 400 mm. Actually deformation is too much small, also it can be neglected.

$$\text{Deformation to length ratio} = \frac{\text{max. deformation}}{\text{total length}} = \frac{0.000073}{400}$$

Deformation to length ratio is too much less than 2% . Main rib of suction arm is in accepted range.

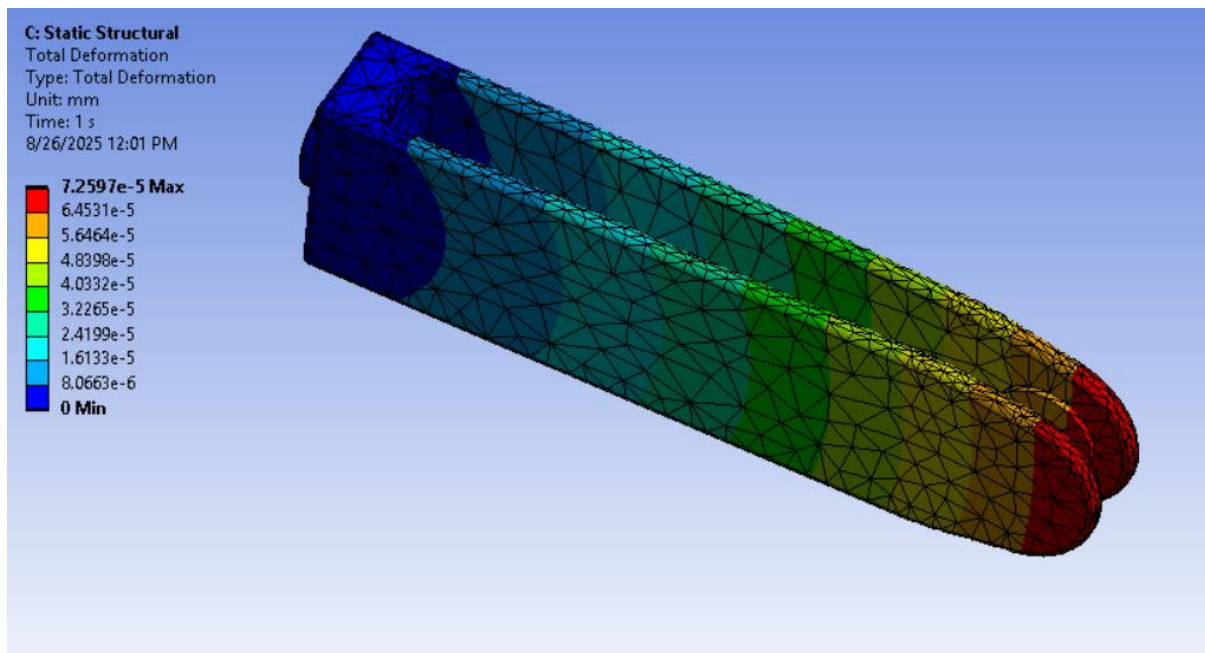


Figure 21: Total deformation of supporting rib

The main rib of suction arm is made from Aluminum and its allowable tensile stress in 70- 80 MPa range. For Ansys simulation gives maximum equivalent stress developed is 0.0084 MPa (figure 22) and stress also not exceeding allowable limit.

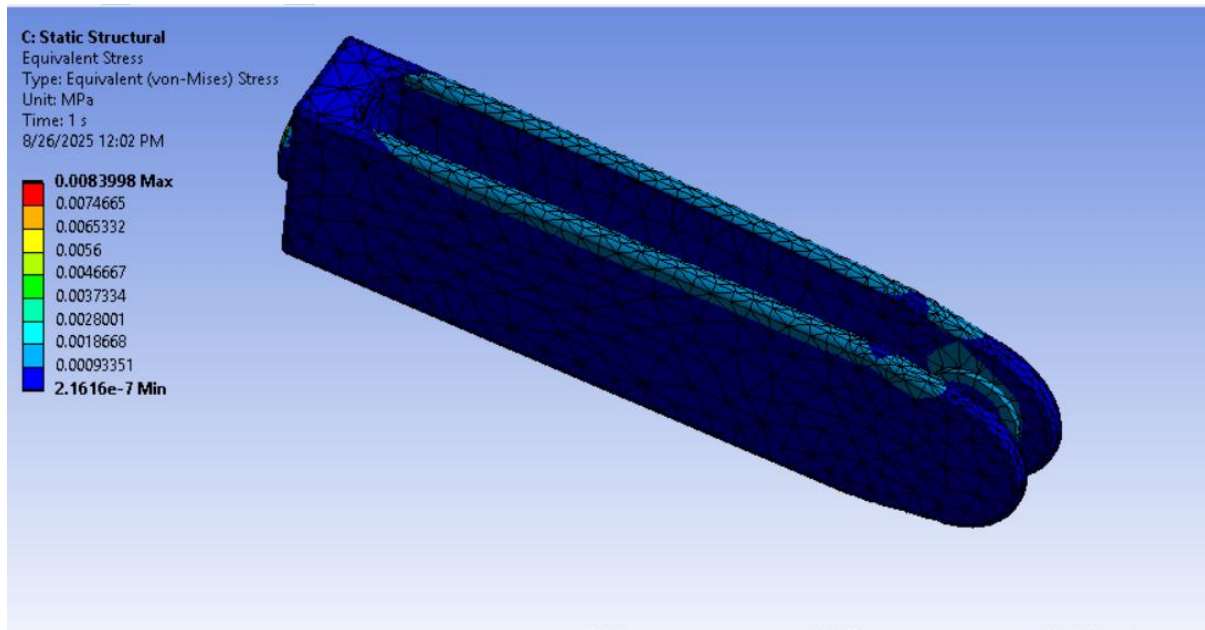


Figure 22: Equivalent stress of supporting rib

Through the engineering analysis and testing of the arm structure, excellent results were obtained in all the individual rib components making up framework. When we examined the equivalent levels of shear stress, i.e. how many pieces of Actual piece of equipment with a broken rib This implies that no rib is overworked and working under the conditions it cannot sustain during regular operation. By simultaneously measuring the extent to which each rib deflects or moves due to the loads applied, we have determined that due to the loads the causes of problems is quite small. When the forces are exerted on the ribs, the movement of the ribs is minimal, but it shows that it is strong enough and rigid enough to serve its role. The same low deformation will also mean that the arm will retain its shape and position during use well without sagging or bending in ways that can affect its performance. Generally, such results indicate to us that the rib structure is performing in its intended way and there are excellent safety margins, and the arm structure is dependable in its anticipated uses.

Adhesive Tank

The adhesive storage tank is a vital part of the tile installation machine and has been specially designed to accommodate and store 1 kilogram (or about 10 Newton of the mass that is attracted by the gravitational force) worth of adhesive material during use. It has a sturdy 10 mm- thick walls all over its body, which provide it with enough strength and durability required to withstand normal working environments. To confirm the structural soundness and safety of such a design, finite element analysis was thoroughly done using ANSYS simulation software. All safety requirements, along with the engineering standards, are satisfied through the

simulation results that prove that the tank design is suitable. The analysis shows that the 10-millimeter thickness of walls gives adequate structural support to carry the full 1-kilogram force of adhesive used without risk of serious stress concentrations, dangerous deformation or possible failure mechanisms. The simulation results in ANSYS reveal that the stresses are well below the acceptability limit, deformation is unreasonable and the design also employs reasonable safety factors. This establishes the assurance that the adhesive tank can adequately meet the requirements of the intended use, during the anticipated lifetime of the tile installation machine, and assures that the operator of the machine feels safe and secure in carrying out an adhesive handling process within the confines of the adhesive tank system design herein described and illustrated.

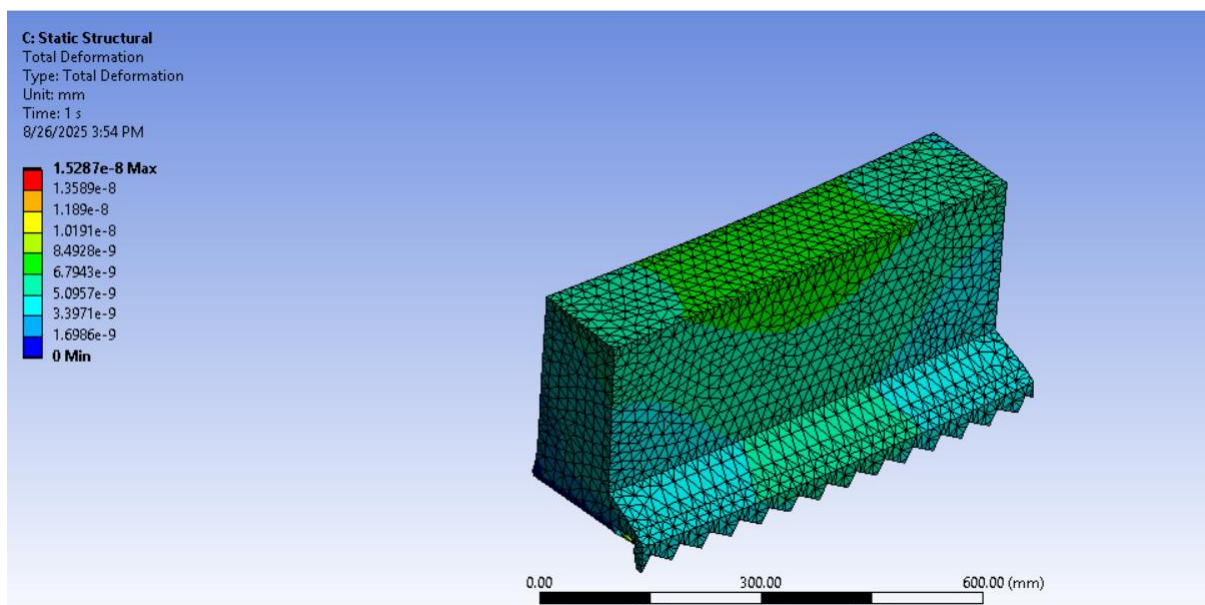


Figure 23: Total deformation of adhesive tank

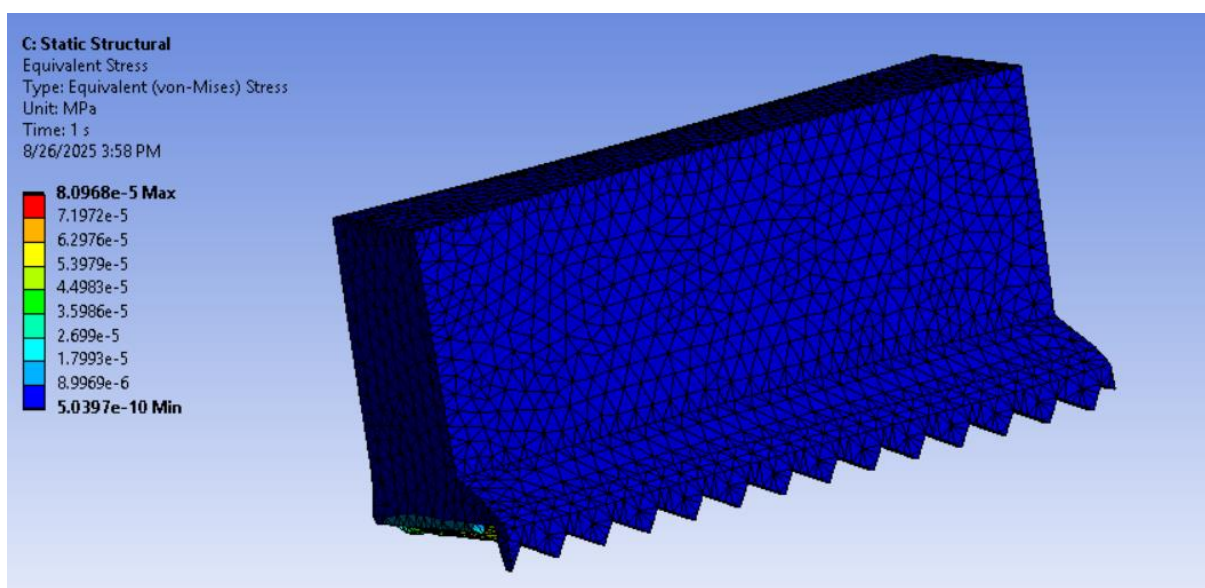


Figure 24: Equivalent stress of adhesive tank (1)

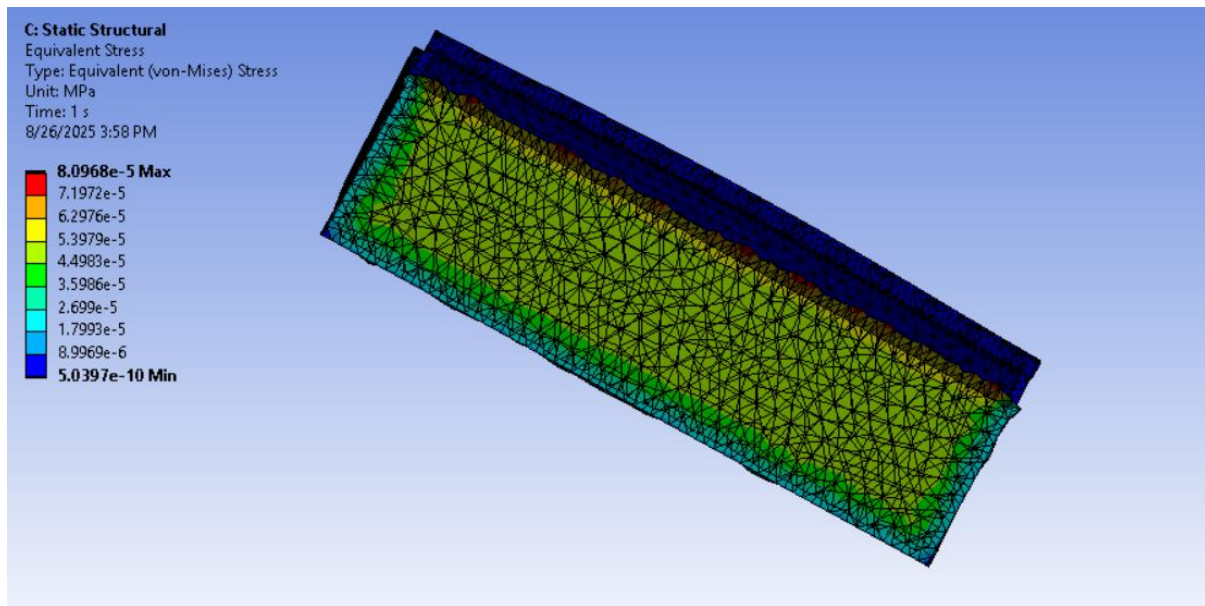


Figure 25: Equivalent stress of adhesive tank (2)

Frame

According to the results of structural analysis provided in the figure below, the structural framework design appears to be appropriately designed and all targeted engineering requirements and safety requirements are satisfied. The frame, made of high grade structural steel material, has stress levels and deformation behavior that are adequately far under the limit of acceptability in standards and design codes associated with industry practices. It is as a result of these thorough simulation findings that the structural frame design has been adopted as a go project. The structural steel construction allows the required strength, durability, and reliability to ensure appropriate structures are achieved to withstand the intended application. All the important stress conditions, members which are subject to load, and the details of connections are checked with flexibility of the finite element analysis to reveal the safe and reliable work of the structure at least during the countable period of the service term. The structural frame has been approved to ensure that it could withstand the structural loads and all the other loading conditions anticipated in the building design. The main building material used, which is structural steel, will enable the adoption of structural behavior that can be projected in light of diverse loading scenarios, satisfying consistent mechanical properties, and structural durability over the long-term period through the use of this approved framework design.

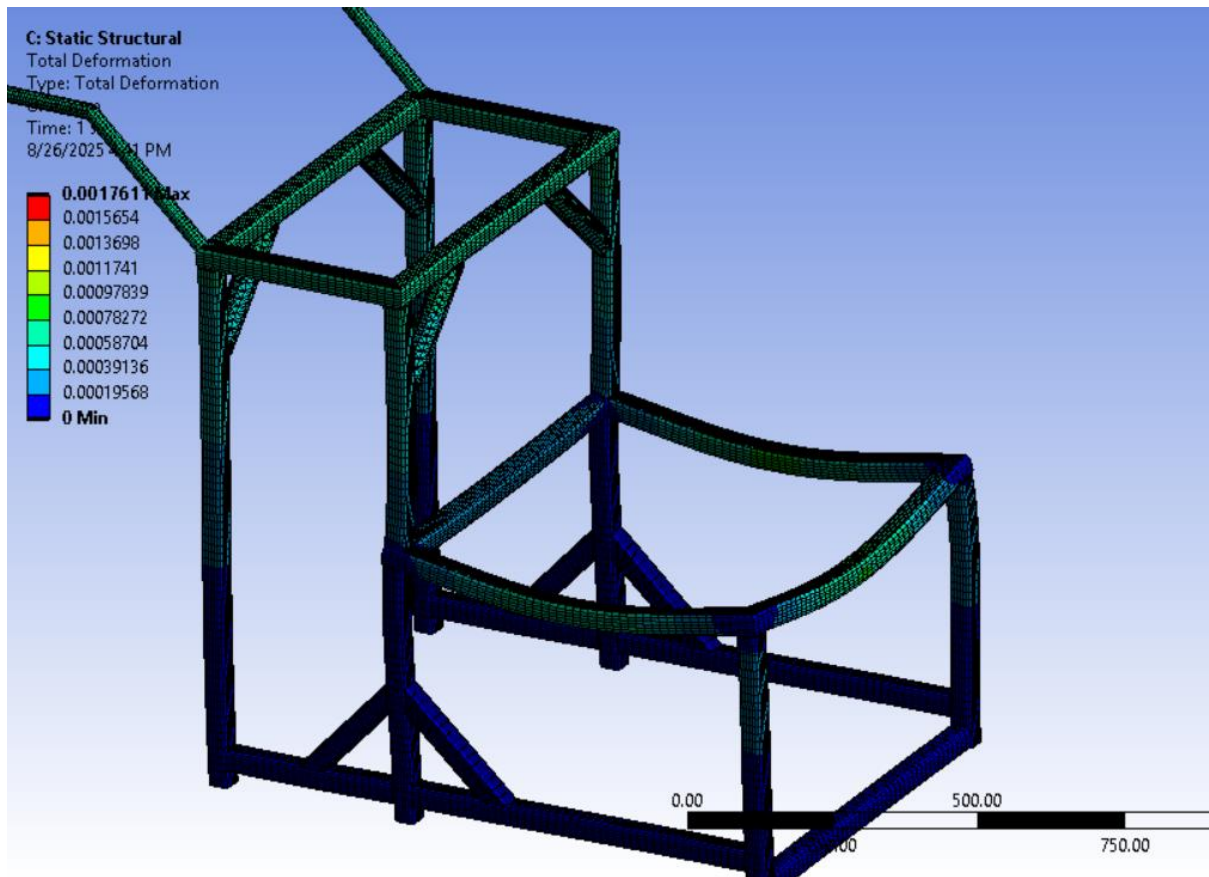


Figure 26: Total deformation of metal frame

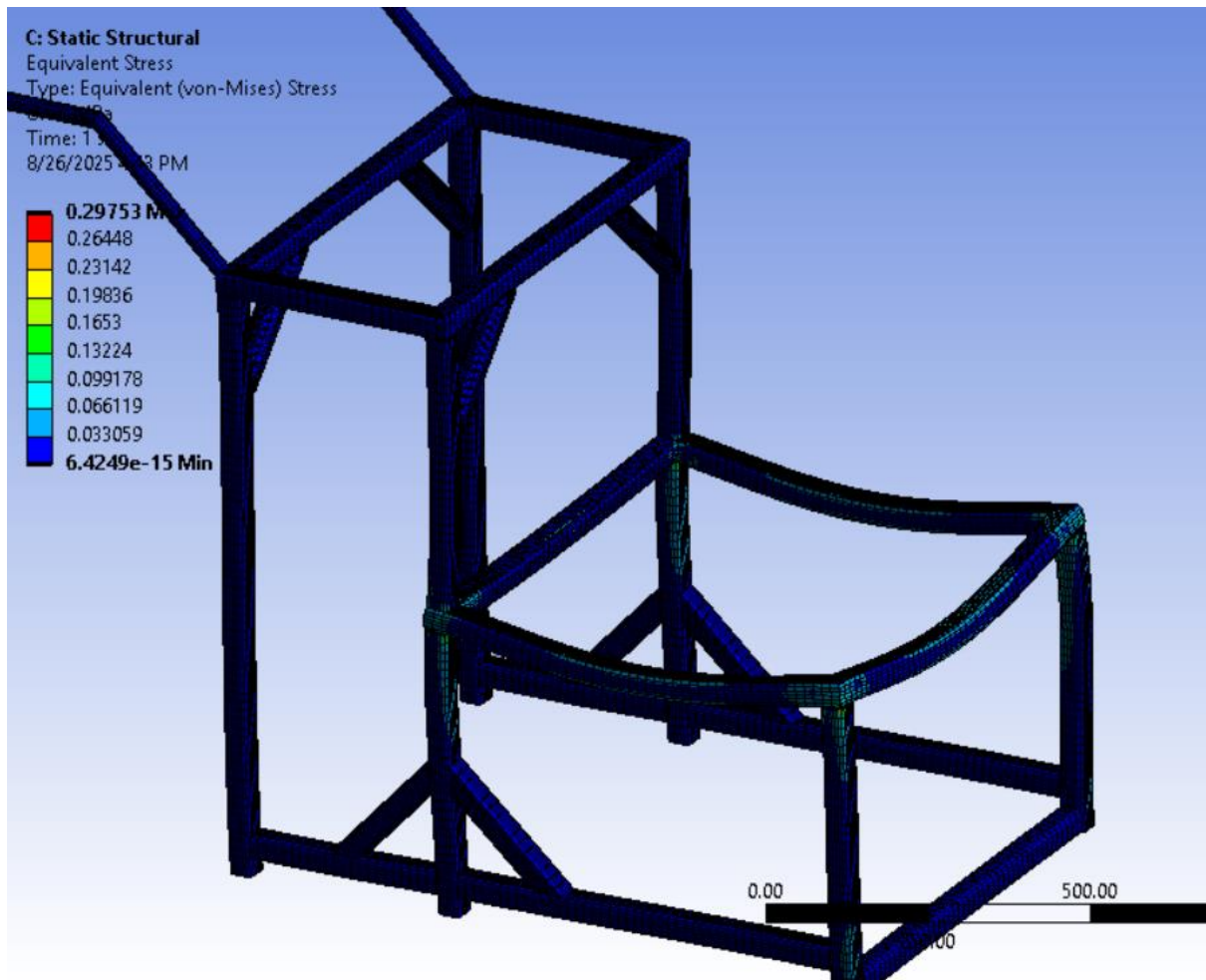


Figure 27: Equivalent Stress- frame

Conclusions and Recommendations

The prototype Mechanically Updated Semi-Automatic Tile Installing Machine has been conceived and designed successfully and validated as a workable option to stop the inefficiencies and inconsistencies of the manual method of installing tiles. The project has also shown that it is possible to design and fabricate mechanical subsystems mechanics (such as the rigid frame, adhesive spreading system, tile placement arm, pneumatic suction hand, and spirit leveling mechanism) which largely enhance accuracy, efficiency and convenience to the user whilst remaining within a reasonable budget.

Design and Analysis

CAD modeling and simulation and FEA were used to establish the structural integrity of key structures and points of loads to ensure that they are safe and durable under anticipated working conditions.

Subsystem Functionality

The spreading system of the adhesive provided an equal thickness of adhesive with the variations of less than or equal to 1 mm and the suction arm module allowed a safe handling of the tiles up to 600 cmx600 mm without scratches.

Precision and Performance

The prototype testing confirmed that the tile placement error was within 1 mm, tile spacing was consistently maintained at 3-5 mm and that tile leveling error was <1 mm over 1 m. The speed of installation rose to an average of 40-50% of that of manual tiling.

User validation

Testing with small contractors and doing it yourself, customers confirmed that it was easy to operate, portable and eliminated labor, thus making it suitable for a non-professional.

Market Relevance

With the lower end of handheld tools being unacceptably low in price and the higher end of robotized tiling systems being prohibitively expensive, the machine can provide a semi-automated application to all but impossibly small-scale buyers.

Overall, the project has demonstrated that semi-automatization can be the way to modernize the tile laying without the price, size, and integrating difficulties of fully robotized systems. The designed machine achieves its aims of accuracy, saving labor and cost-effectiveness to a small contractor and domestic applications.

Recommendations

It is recommended that additional work and refinements be made to improve performance, reliability and commercial viability whilst being mindful of meeting the initial project requirements stated.

Prototyping, fine tuning and on-ground testing.

Perform prolonged usability tests on the ground under the real conditions to demonstrate reliability over the long term and in ambiguous conditions (dust, unsmooth floors, humidity). Strengthen robotic arm joints and the hinges to reduce the levels of vibrations in case of extended usage.

Enhanced Automation

Combine low-cost microcontrollers to closed loop controllers and control semi-automatic tile position adjustments and tile leveling corrections.

Install digital sensors to enhance accuracy and simultaneously diminish human supervision.

Adhesive System Enhancement

Learn about motor-driven or pump-aided adhesive dispensing to dispense with more consistency and at a lower risk of clogging.

Investigate lightweight composite or polymer tanks to minimize the chassis weight.

Ergonomics, and Portability

Optimize chassis design to provide the lowest possible weight and maintain rigidity potentially with use of aluminum or composites.

Make them be folded or modular in order to travel and store them more easily.

Cost and Manufacturing Optimization

Add value to the Bill of Materials (BOM) with focus on standardized parts to reduce cost.

Evaluate low volume production processes (CNC machining, sheet metal forming, additive manufacturing of the suction hands).

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